4. Hydroboration:

Diborane reacts with an alkene to form trialkyl borane which on treatment with H_2O_2 in presence of NaOH gives an alcohol. (Refer reactions of diborane) The overall reaction is hydration of an alkene. This reaction yields an anti-Markownikoff's product.

$$6CH_3-CH=CH_2 + B_2H_6 \longrightarrow \underset{\text{Tripropylborane}}{2(CH_3-CH_2-CH_2)_3B}$$

$$(CH_3-CH_2-CH_2)_3B + 3H_2O_2 \xrightarrow{OH^-} \underset{\text{propan-1-ol}}{3CH_3-CH_2-CH_2-OH} + B(OH)_3$$

5. Reduction of carbonyl compounds:

Reduction of aldehydes/ketones with $LiAlH_4$ in the presence of solvents like THF (Tetrahydrofuran) followed by hydrolysis gives alcohols. Unlike other reducing agents such as Raney Ni, Na-Hg/H_2O, the lithium aluminium hydride does not reduce the carbon–carbon double bond present in unsaturated carbonyl compound and hence it is a best reagent to prepare unsaturated alcohols.

Examples

$$\underset{\text{acetone}}{CH_3-\overset{\overset{CH_3}{|}}{C}=O} \xrightarrow[\text{ii) } H_2O]{\text{i) } LiAlH_4} \underset{\text{propan-2-ol}}{CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}}-H}$$

$$\underset{\text{Benzoic acid}}{C_6H_5-COOH} \xrightarrow[(ii)H_2O]{(i)LiAlH_4} \underset{\text{Phenyl methanol(Benzyl alcohol)}}{C_6H_5CH_2OH}$$

$$\underset{\text{Ethyl ethanoate (ethyl acetate)}}{CH_3COOCH_2CH_3} \xrightarrow[(ii)H_2O]{(i)LiAlH_4} \underset{\text{Ethanol}}{2CH_3CH_2OH}$$

$$\underset{\text{crotonaldehyde (but-2-enal)}}{CH_3-CH=CH-CHO} \xrightarrow[(ii)H_2O]{(i)LiAlH_4} \underset{\text{crotyl alcohol (but-2-en-1-ol)}}{CH_3-CH=CH-CH_2OH}$$

When two or more functional groups are present in a molecule a less vigorous sodium borohydride is used as a reducing agent to reduce the more reactive group. For example, if a compound contains both carbonyl and carboxyl group, it preferentially reduces the carbonyl group.

$$RCOCH_2CH_2COOH \xrightarrow[H_3O^+]{NaBH_4} RCHOH-CH_2CH_2COOH$$

4-alkyl- 4- hydroxybutanoic acid

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